L4.3b Which Method?

Disks, washers and shells — what you need to know

CHOOSING THE METHOD

① **Which variable are the curves written in?** $y = f(x)$: you want dx. $x = g(y)$: you want dy. Pick the one that needs no solving for the other variable.

② **Draw a thin rectangle of that thickness:** dx gives an upright rectangle, dy a sideways one.

③ **Compare it with the spin axis: perpendicular** ⇒ disks or washers. **Parallel** ⇒ shells.

④ **Check:** does the rectangle run between the same two curves across the whole region? Can you integrate what you get? If not, switch variables and go back to ②.

	curves in x: dx, upright rectangle	**curves in y:** dy, sideways rectangle
axis horizontal (x-axis, $y = k$)	disks or washers	shells (about the x-axis)
axis vertical (y-axis, $x = k$)	shells	disks or washers

DISKS AND WASHERS

① **Draw a representative disk or washer:** a thin rectangle **perpendicular** to the spin axis, and the dV it sweeps. Label $r(x)$, or R and r.

② **Path and limits:** along the spin axis, dx or dy.

③ **Radii:** measured from the axis, in the variable of the path.

④ **Chunk:** $dV = \pi\, r(x)^2\, dx$, or $\left(\pi R^2 - \pi r^2\right) dx$.

$$V = \int_{\text{path}} dV$$

SHELLS

① **Draw a representative shell:** a thin rectangle in the region, **parallel** to the spin axis. Spun, it stands on the ring $dA = 2\pi r\, dr$.

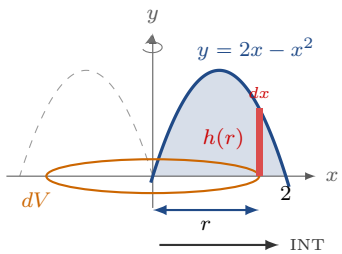
② r **is the distance** from the spin axis.

③ $h(r)$ **is the length** of the rectangle inside the shaded region, written in r.

④ **Limits:** the inner and outer r.

$$V = \int_{\text{inner}}^{\text{outer}} h(r)\, dA$$

ONE SOLID, TWO METHODS



Ex. Rotate the region under $y = 2x - x^2$, $0 \leqslant x \leqslant 2$, about the y-axis.

Choose. The curve is $y = f(x)$, so dx: an upright rectangle, **parallel** to the y-axis. Shells.

Shells. $r = x$, $h(r) = 2r - r^2$, inner 0, outer 2:

$$V = 2\pi \int_0^2 2r^2 - r^3\, dr = 2\pi\left[\tfrac{2}{3}r^3 - \tfrac{1}{4}r^4\right]_0^2 = 2\pi\left(\tfrac{16}{3} - 4\right) = \boxed{\tfrac{8\pi}{3}}$$

Washers instead. Slices perpendicular to the y-axis need x in terms of y. Po-Shen Loh on $x^2 - 2x + y = 0$: sum 2, product y, so $x = 1 \pm u$ with $1 - u^2 = y$, and $x = 1 \pm \sqrt{1-y}$:

$$R(y) = 1 + \sqrt{1-y} \qquad r(y) = 1 - \sqrt{1-y} \qquad R^2 - r^2 = 4\sqrt{1-y}$$

$$V = \pi \int_0^1 4\sqrt{1-y}\, dy = 4\pi \cdot \tfrac{2}{3} = \boxed{\tfrac{8\pi}{3}}$$

The same number, with more work.

WATCH OUT

✗ ~~shells about the y-axis in dy~~ → **shells about the y-axis use dx: $r = x$, the rectangle is upright**
✗ ~~$V = \int 2\pi r \cdot \pi r^2 \, dr$~~ → **disks square the radius, shells multiply by $2\pi r$ — never both**
✗ ~~washers about the y-axis in x~~ → **washers about the y-axis use dy: write x in terms of y**